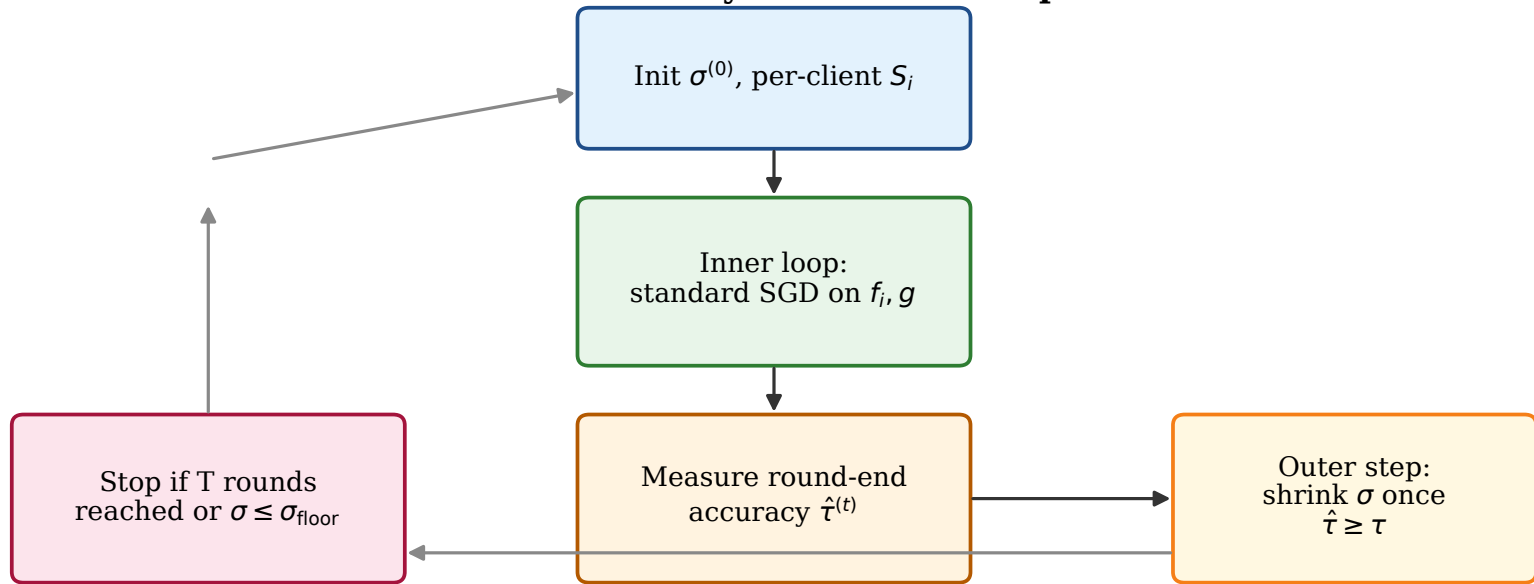


Bi-level utility-bounded DP loop



$$\sigma^{(t+1)} = \max(\sigma_{\text{floor}}, \rho \cdot \sigma^{(t)}) \text{ once } \hat{\tau}^{(t)} \geq \tau$$